

Tasnim Amara, B.A.SC. (2030)

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SKILLS

Programming	Proficient in Python, Java, MATLAB, HTML/CSS
Engineering Software	Skilled with using Microsoft Office, SIMULINK, SolidWorks, Altium Designer, Traxmaker PRO
Hardware/Laboratory	Well-versed in Micropipetting, Electroplating, Material Performance Testing, Literature Review, PIC Microcontrollers, Arduino, PCB Assembly & Troubleshooting
Soft Skills	Experience with Project Coordination, Technical Documentation, Team Collaboration

WORK EXPERIENCE

Undergraduate Research Assistant **Dec 2025 - Present**
Multi-Scale Additive Manufacturing Lab, Kitchener, ON

- Supporting an advanced materials research project developing next-generation sensors for structural health monitoring in high-stress industrial applications.
- Assisting in fabrication and testing of specialized sensor components using precision thin-film deposition and etching techniques (including Astrojet system operation).
- Conducting materials science research focused on developing thermally stable composite formulations capable of withstanding extreme operating conditions.
- Performing laboratory analysis and data collection to evaluate material performance under simulated stress environments.

Summer Reading Club Facilitator **Jun 2025 - Aug 2025**
Waterloo Public Library, Waterloo, ON

- Acted as the welcoming face for families, managing registrations and inquiries, directly increasing program retention through personalized interactions.
- Collaborated with a team to plan and execute a diverse schedule of 4 weekly activities, demonstrating strong project coordination and time-management skills.
- Coordinated a large-scale end-of-summer party for over 200 attendees, improving logistical planning and teamwork abilities.

PROJECTS

Webcam-Controlled Robot *Arduino, MATLAB/SIMULINK, Computer Vision*
Developing a webcam-controlled rover for path following, object manipulation (forklift attachment), and obstacle avoidance. Programming Arduino Nano 33 IoT for real-time motor control and integrating computer vision algorithms in MATLAB for visual navigation. Building mechanical structure, integrating servo motors/drivers, and troubleshooting hardware/circuit implementation.

Firefighter Flame Extinguisher Robot *Microcontrollers, PCB Design, Troubleshooting*
Built an autonomous fire-extinguishing robot that detects flames using IR/ultrasonic sensors and navigates to the source. Designed custom PCBs from schematic using Traxmaker PRO and systematically troubleshooted circuits with multimeter to ensure full functionality. Programmed PIC16f18875 microcontroller with autonomous logic in PICBasic Pro to identify flames and activate suppression. Developed pathfinding routines using ultrasonic/IR sensors for 100% consistent flame detection.

Triboelectric Nanogenerator (TENG) *SolidWorks, 3D Modeling, Prototyping*
Designed and modeled a vortex-whistle TENG in SolidWorks to convert wind energy into electricity for portable lighting. Prototyped via 3D printing, optimizing materials for triboelectric output (0.4V measured). Collaborated in a team to iterate design and prepare technical documentation.

EDUCATION

University of Waterloo Expected 2030
B.A.Sc. Honours Nanotechnology Engineering

- Awarded the University of Waterloo President's Scholarship for outstanding academic performance

Last updated: February 4, 2026